

1. Administrative & Study Identification

**Full Study Title:** A Phase II, Randomised, Adaptive, Open-Label Platform Trial To Evaluate The Efficacy And Safety Of Multiple Combination Therapies In Participants With Chronic Hepatitis B  
**Short Title/Acronym:** Piranga **Protocol Number:** 2019-002086-35 **NCT Number:** NCT04225715 **Sponsor Name:** Hoffmann-La Roche  
**Investigational Product:** Xalnesiran (small interfering RNA) / Ruzotolimod / Pegylated interferon alfa-2a **Phase of Development:** Phase II **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate the efficacy of xalnesiran, with or without an immunomodulator, in achieving hepatitis B surface antigen (HBsAg) loss in participants with chronic hepatitis B (HBV) infection. **Secondary Objectives:** To evaluate the percentage of patients with HBsAg seroconversion, hepatitis B e-antigen (HBeAg) loss, and HBV DNA lower than the limit of quantification (LLOQ) up to 96 weeks, as well as to assess pharmacokinetic and safety outcome measures.

**2.2 Background & Rationale** Standard care for chronic HBV, which includes pegylated interferon therapy and lifelong nucleoside or nucleotide analogue (NA) therapy, rarely leads to a functional cure (which occurs in only up to 7% of patients). Xalnesiran is a novel N-acetyl-d-galactosamine-conjugated synthetic double-stranded small interfering RNA molecule that targets the S conserved region of the HBV genome to silence multiple transcripts. This study is conducted to determine if combining this new antiviral agent (to reduce the antigenic load) with immunomodulators (to restore the dysregulated immune response) can significantly improve functional cure rates.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional (Platform Trial)  
**Control Type:** Active Comparator (Nucleoside or nucleotide analogue alone) **Blinding:** Open-label **Randomization:** Randomized (Adaptive)

**3.2 Planned Enrollment & Duration Target** **SN\$:** 159 participants  
**Number of Sites:** Multicenter **Estimated Study Start:** Early 2020  
**Estimated Primary Completion:** 19-Jul-2024

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Confirmed diagnosis of chronic Hepatitis B (HBV) infection. 2. Virologic suppression achieved and maintained with nucleoside or nucleotide analogue (NA) therapy. 3. Preserved liver function without significant fibrosis or cirrhosis.

**4.2 Exclusion Criteria** 1. Co-infection with Hepatitis C Virus (HCV) or Human Immunodeficiency Virus (HIV). 2. Presence of significant liver fibrosis or clinical evidence of cirrhosis. 3. Any unstable medical disorder that can impact participant safety or study outcomes.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Participants are randomized into five groups: \* Group 1: Xalnesiran 100 mg alone \* Group 2: Xalnesiran 200 mg alone \* Group 3: Xalnesiran 200 mg + 150 mg ruzotolimod \* Group 4: Xalnesiran 200 mg + 180 µg pegylated interferon alfa-2a \* Group 5: NA therapy alone **Route of Administration:** Subcutaneous (Xalnesiran, ruzotolimod, pegylated interferon alfa-2a) and Oral (NA therapy). **Duration of Treatment:** Up to 48 weeks of treatment phase, followed by a 48-week post-treatment follow-up phase.

**5.2 Concomitant & Prohibited Therapy Permitted:** Background nucleoside or nucleotide analogue (NA) therapy to maintain virologic suppression. **Prohibited:** Concurrent investigational therapies or alternative systemic immunomodulators outside of the assigned study arm.

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days)        | Key Activities/Procedures                                                                               |
|--------------|----------------------|---------------------------------------------------------------------------------------------------------|
| Screening    | Up to 8 weeks        | Informed Consent, Medical History, Liver function tests, HBsAg baseline levels                          |
| Baseline     | Day 0                | Randomization, First Dose of Xalnesiran/Immunomodulator                                                 |
| Interim      | Weeks 12, 24, 36, 48 | Safety Labs (ALT monitoring), Efficacy Assessment, PK/PD tracking                                       |
| End of Study | Week 96              | Final HBsAg seroconversion assessment, Exit Interview, Safety follow-up (24 to 48 weeks post-treatment) |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Proportion of patients achieving HBsAg loss at 24 weeks after the end of treatment. **Measurement Method:** Central Laboratory Analysis for HBsAg levels (evaluating thresholds such as < 1000 IU/mL and < 100 IU/mL).

**7.2 Secondary & Exploratory Endpoints** 1. Percentage of patients with HBsAg seroconversion. 2. Percentage of patients with HBeAg loss. 3. Percentage of patients with HBV DNA lower than the limit of quantification (LLOQ). 4. Pharmacokinetic and safety outcomes (including incidence of Grade 3/4 AEs such as ALT elevations).

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring during the 48-week treatment and 48-week follow-up phases, with specific focus on Grade 3 or 4 adverse events (e.g., elevated alanine aminotransferase levels), which were noted to be frequent in combination therapy arms. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours) for any life-threatening events or those requiring hospitalization. **Data Monitoring Committee (DMC):** External independent committee to evaluate safety and efficacy signals due to the adaptive platform design.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) population for all efficacy endpoints; Safety Set for all participants who received at least one dose of the study medication. **Sample Size Justification:** Sample size of  $N=159$  allocated adaptively across 5 arms (approx. 30-35 per arm) to provide sufficient power to detect a meaningful clinical signal in HBsAg loss compared to the standard of care. **Handling of Missing Data:** Missing virologic response data at evaluation time points are generally treated as non-response (failure to achieve HBsAg loss).

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol compliance and safety tracking, particularly regarding hepatic flare management. **Audit Trail:** Continuous eCRF locking and query resolution handled by the sponsor's data management team.

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## 11. Study Identifiers & Governance

**Primary ID:** 2019-002086-35 **Registry Number:** NCT04225715 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** Piranga **Official Regulatory Title:** A Phase II, Randomised, Adaptive, Open-Label Platform Trial To Evaluate The Efficacy And Safety Of Multiple Combination Therapies In Participants With Chronic Hepatitis B

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## 12. Clinical Framework (Hepatitis B Specific)

**Primary Condition:** Chronic Hepatitis B (CHB) **Diagnosis Threshold:** Positive HBsAg with virologic suppression on NA therapy **Medical Methodology:** siRNA gene-silencing therapy combined with immune response modifiers (immunomodulators) **Optimization Target:** Functional cure (Durable HBsAg loss and seroconversion)

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## 13. Study Procedures & Methodology

**Management Pathway:** Sequential or combination administration of RNA interference therapy and immunomodulators. **Education Protocol (HCP):** Training on administration of subcutaneous agents and strict monitoring protocols for hepatotoxicity (ALT flares). **Education Protocol (Patient):** Education on reporting adverse events and maintaining adherence to background NA therapy. **Quality Audit Mechanism:** Centralized laboratory validation of HBsAg and HBV DNA limits.

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Up to 96 weeks (48 weeks treatment + 48 weeks post-treatment follow-up) **Onsite Visit Cadence:** Regular intervals through Week 48, with critical follow-ups at Week 72 (24 weeks post-treatment) and Week 96 (48 weeks post-treatment). **Remote Monitoring Frequency:** Between onsite visits for AE reporting. **Checklist Review Interval:** Every 12 weeks for virologic milestones.

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|------|------|------|------|
| Role                     | Name   | Signature | Date          | :--- | :--- | :--- | :--- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |      |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |      | Lead |      |      |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |      |      |      |      |